

Dynamic Planet Test

Columbia Invitational

Team #: _____

School Name: _____Key_____

Participants: _____ &

_____

Be sure to put all of your answers on the answer sheet.
The question packet will not be graded.

1. Match the terms with their appropriate definitions (1 pt each)
   - Word Bank : Karst spring, karst canyon, dry valley, grike, swallet, sinkhole, polje, solution tube
     a. _____**Sinkhole** –a topographically closed depression that is circular or elliptical in shape and with steep to vertical sidewalls. (Also called a 'doline' in European texts),
     b. _____**Swallet** – a point where a stream of any size sinks underground, forming (in some cases) a cave entrance (Figure 12.2.4),
     c. _____**Dry valley** – a linear valley that did (or occasionally does) contain a stream,
     d. _____**Karst canyon** – a steep sided canyon in karst with distinctive surface erosional features (e.g., scalloping),
     e. _____**Karst spring**– a site where an underground stream emerges from the karst bedrock and is sometimes the site for a cave entrance,
     f. _____**Polje**– a large flat bottomed karst depression with water flowing at the bottom,
     g. _____**Grike** – a linear, narrow, and deep slot formed by dissolution along a pre-existing fracture in bedrock, and
     h. _____**Solution tube** – a circular or elliptical, steeply inclined tube formed by dissolution.

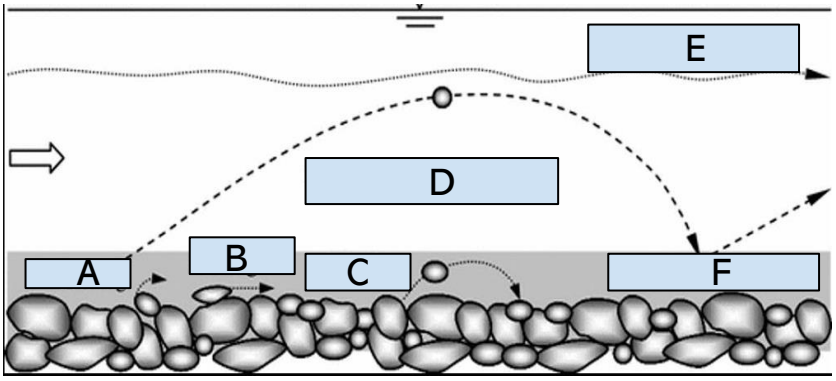


Figure 1: Sediment Transport

2. Letters A through C represent modes of sediment transport in a river. Letters D through F represent types of sediments being transported in a river. Label them using the word bank below. (1 pt each)

wash load, bed load, suspended load, sliding, rolling, saltation

a. _____ **rolling**
b. _____ **sliding**
c. _____ **saltation**
d. _____ **suspended load**
e. _____ **wash load**
f. _____ **bed load**

3. What relationship does the amount of canopy coverage have with direct throughfall onto the soil? (1pt)
    A. A direct relationship
    B. An inverse relationship
    C. No significant relationship
    D. Relationship depends on vegetation type

4. Around which value is the typical pH of normal rain? (1pt)
    A. 3
    B. 5
    C. 7
    D. 9

5. Which of these tidal wetland types matches the description "Tidal water covers the land surface less often than daily"? (1pt)
    A. Subtidal
    B. Irregularly exposed
    C. Regularly flooded
    D. Irregularly flooded

6. Select all of the following that the thawing of permafrost would cause. (1 pt each)
    A. Decrease the permeability of the subsurface
    B. Not affect the permeability of the subsurface
    C. Increase the permeability of the subsurface
    D. Would cause the size of a small lake to decrease
    E. Would cause the size of a small lake to increase

7. Lake turnover occurs because of the weakening of. (1 pt)
    A. Halocline
    B. Epicline
    C. Trade winds
    D. Thermocline

8. A tectonic lake is most likely to form in a: (1 pt)
    A. Horst
    B. Graben
    C. Transpression
    D. Transform fault

9. Which of the following is not a characteristic of a bog? (1 pt)
   A. Is also known as a muskeg
   B. Has acidic water
   C. Has a low peat content
   D. Most water received through precipitation

10. Which of the following can support trees? (1 pt)
   A. Bogs
   B. Marshes
   C. Fens
   D. Swamps

11. Which of the following are found near the ends of glacial moraines? (1 pt)
   A. hemi-marsh
   B. **seep**
   C. swamp
   D. emergent marsh
   E. none of the above

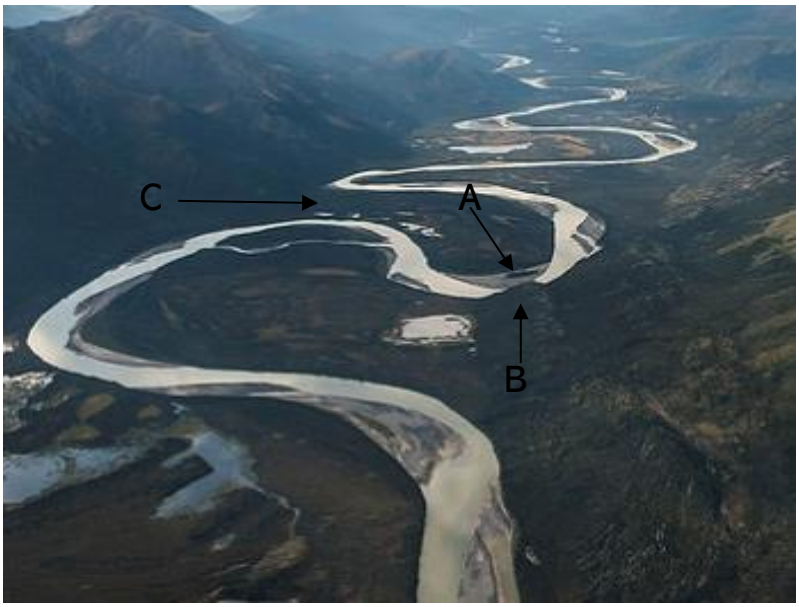


Figure 2: Channel Types for questions 12 - 19

12. What type of channel is this? (1 pt) **Meandering**
13. What is the name of this specific famous river? (2 pt) **Alatna River**
14. A is pointing to a sandy bank along the concave portion of the river. What is this called? (1 pt) **Point Bar**

15. B points to the bank opposite of A, on the convex side of the river. What is it called? (1 pt) **Cutbank**
16. Answer with lower or higher. Flow velocity is _____ around point A and _____ around point B. (2 pt) **lower, higher**
17. C points to the narrow piece of land between the curves of the stream. What is it called? (1 pt) **Neck**
18. Given a long time span, if the small channel continues to grow, what will happen to the original curve of the river that it bridges, and what kind of feature will it form? (3 pt)
    When the small channel grows, it will divert more of the river's flow, and sand/silt will build along the channel's banks until the original curve is isolated, forming an oxbow lake. (some reasonable explanation with oxbow lake mentioned)
19. Generally, the slope of the river will cause the curves to migrate _____. (1 pt)
    - upstream
    - **downstream**

20. Which of the following are oxbow lakes most associated with? (1pt)
- A. Braided channels
- B. Anastomosing channels
- C. **Meandering channels**
- D. Straight channels

21. Accumulations of sediment formed by the movement of water on the sides of a flood plain are called ____ (1pt)
- A. Backswamp
- B. Deltas
- C. Alluvial fans
- D. **Natural levees**

22. In the profile of a typical river, how does the gradient change from head to mouth? (1pt)
- A. Does not change
- B. Varies constantly
- C. Increases
- D. **Decreases**

23. How far can large rivers flow into the ocean and maintain freshwater before mixing with the ocean? (1pt)

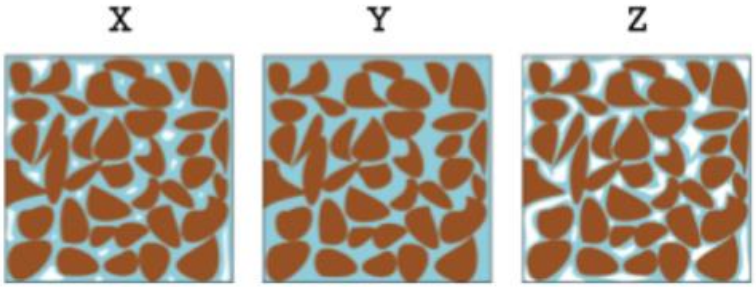
- A. A few meters
- B. Tens of meters
- C. A few kilometers
- D. **Many kilometers**

24. Which of the following acids that natural water contains makes it easy for groundwater to dissolve limestone? (1pt)

- A. Acetic acid
- B. Nitric acid
- C. **Carbonic acid**
- D. Hydrochloric acid

25. Which of the following is the largest freshwater lake by Volume? (1pt)

A. Lake Superior
B. Lake Baikal
C. Lake Tanganyika
D. Lake Victoria



26. Shows illustrations of cross sections of soil at different soil moisture thresholds. Which of the following correctly matches the diagram to the threshold? (1pt)

A. X → Saturation. Y → Field capacity. Z → Permanent Wilting Point.
B. X → Field capacity. Y → Saturation. Z → Permanent Wilting Point.
C. X → Saturation. Y → Permanent Wilting Point. Z → Field capacity.
D. X → Field capacity. Y → Permanent Wilting Point. Z → Saturation.
E. X → Permanent Wilting Point. Y → Field capacity. Z → Saturation.
F. X → Permanent Wilting Point. Y → Saturation. Z → Field capacity.

For questions 27 – 35 match the following drainage patterns with their corresponding description (1pt each)

Word bank:
Trellised, Centripetal, Parallel, Herringbone, Rectangular, Angular, Radial, Barbed, Pinnate

27. This drainage pattern is characteristic of folded mountains, like the Appalachian Mountains. **Trellised**

28. This pattern originates from a hill with flow in all directions; Amarkantak is a notable example. **Radial**

29. Drainage follows a regional slope and is more frequent in newly emerged coastal plains. **Parallel**

30. This drainage pattern forms when streams converge at a point, typically a basin. **Centripetal**

31. This pattern is formed when rocks are eroded along joints or fractures, resulting in clefts or rills used in draining. **Rectangular**

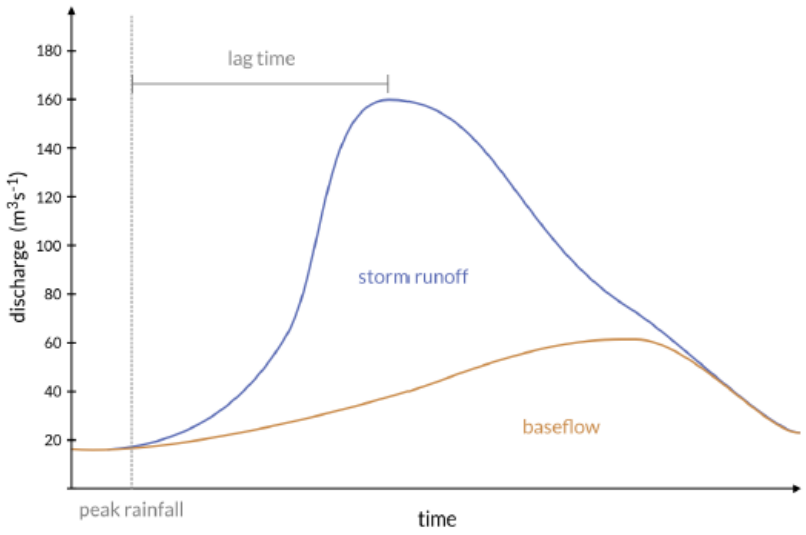
32. This pattern usually develops in mountain-containing regions with steep slopes and valleys, where tributaries and master streams converge. **Herringbone**

33. This drainage resembles the pattern of a leaf. **Pinnate**

34. This drainage pattern originated when tributaries flowed in an opposite direction to the master stream. **Barbed**

35. This drainage pattern is known as the “circular pattern”. **Angular**

For questions 36-39 refer to the following storm hydrograph. Precipitation begins at the start of the x-axis.



36. Further urban development in the watershed upstream would have what result on the hydrograph for a similar precipitation event? (1pt)

A. An increase in lag time
B. A higher peak flow
C. Both A and B
D. Neither A or B

37. How would urbanization most plausibly affect baseflow? (1pt)

A. Baseflow would be higher on average, making the hydrograph broader in width.
B. Baseflow would be higher on average, giving the hydrograph two distinct peaks.
C. Baseflow would be lower on average, making the hydrograph more symmetrical.
D. Baseflow would be lower on average, the hydrograph maintain its overall shape.

38. Which of the following best accounts for the delay from the onset of precipitation to the contribution of storm runoff to stream flow? (1pt)

A. Runoff takes time to reach stream gauges.
B. The ground surface is initially unsaturated.
C. Peak rainfall is reached after precipitation begins.
D. Streamflow consists entirely of baseflow.

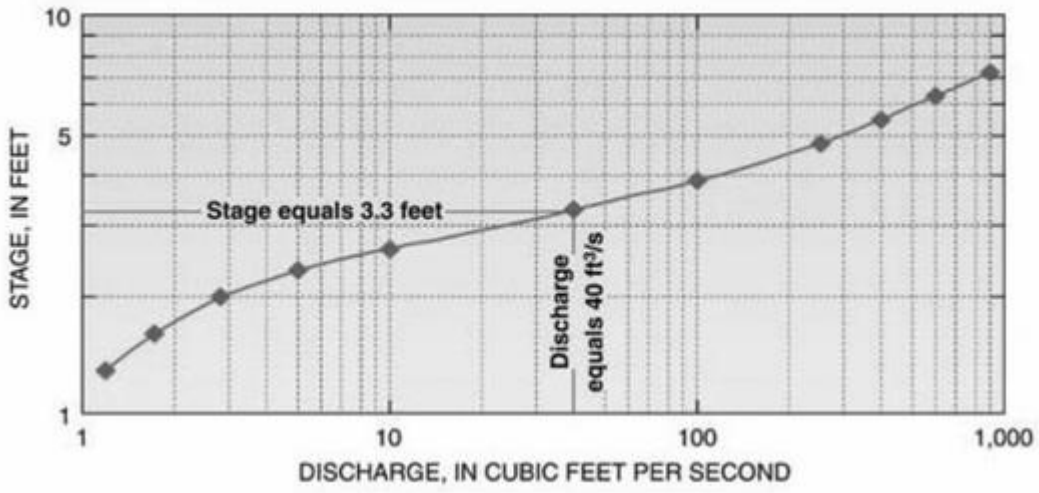
39. At the location of the stream gauge at the time of peak discharge, the cross-sectional area is 85 square meters and the wetted perimeter is 24 m. Calculate the hydraulic gradient at the stream gauge using the Chézy formula, taking the Chézy coefficient to be 20 $m^{1/2}/s$. (1pt)

A. 0.0025 m/m
B. 18.0706 m/m
C. 51.6398 m/m
D. 476.0952 m/m
E. 0.1245 m/m

$V = C^*sqrt(R^*i) = Q/A$

$i = (Q/AC)^2/R = (160\ m^3/s/((85\ m^2)(20\ m^{1/2}/s)))2/(85\ m^2/24\ m) = 0.0025\ m/m$

For questions 40 and 43 use the chart below.



40. The above chart only applies to some rivers where stage is the main cause of discharge variance. What features or phenomena in rivers would the stage-discharge chart not apply to? Select all that apply. (1pt each)

a. **tides**
b. **low slope**
c. **density currents**
d. none of the above

41. Using the above stage-discharge chart, what volume of water, in cubic feet, would pass through a 1x1 cross section of the river at a stage of 5 feet after 10 seconds? (3 pt)

(300 ft^3/s * 10 seconds) = 3000 ft^3

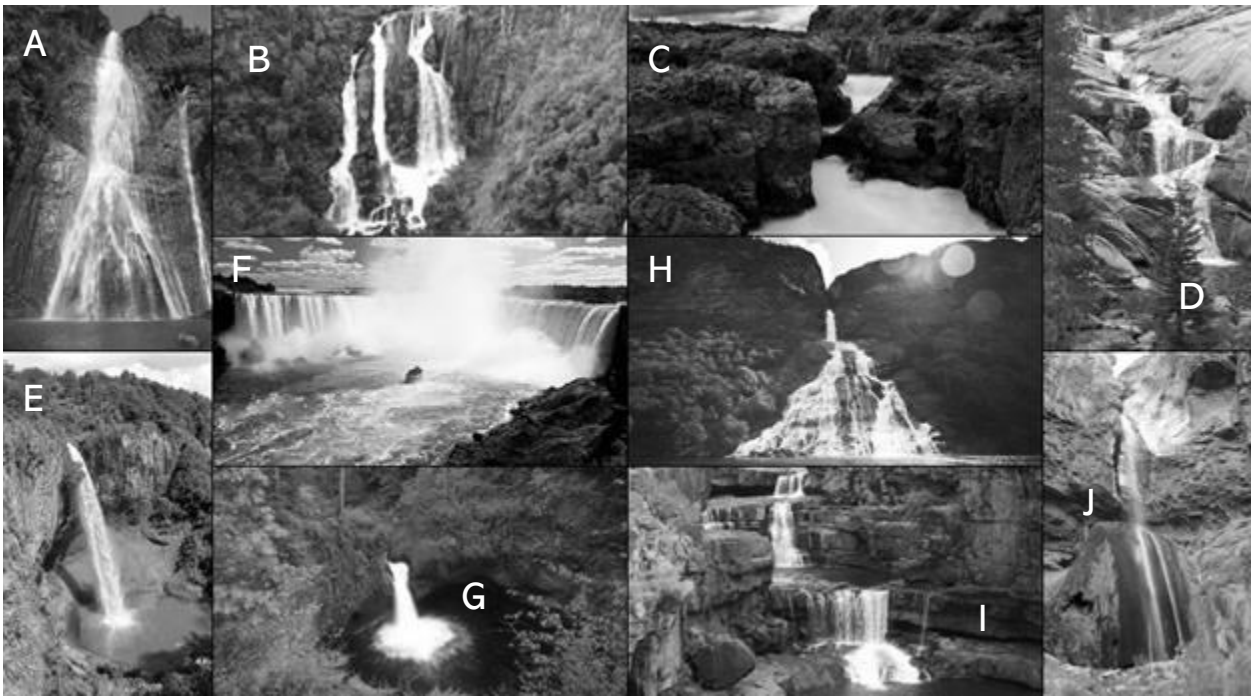
42. Oxbow lakes can be found in valleys with which type of central river? (1pt)

- Young
- **Old**
- Mature
- Adolescent

43. Pocosins are examples of (1pt)

- Rheotropic environments
- Minerotrophic environments
- **Ombrotrophic environments**
- Phreatotrophic environments

Use the below diagram of waterfalls for question 44



44. Waterfalls come in many categories. Assign each of these waterfalls with a category from the word bank. There are no repeats. (1pt each)

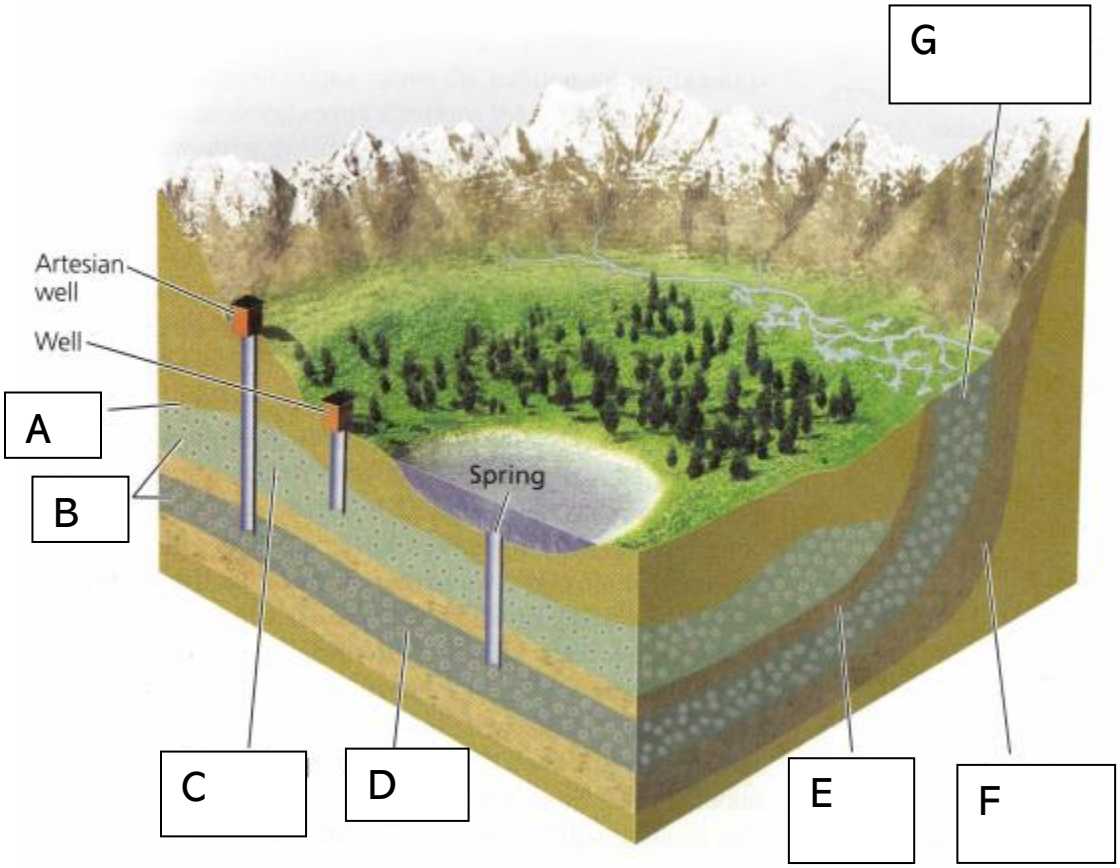
Ribbon, Horsetail, Plunge, Punchbowl, Block, Fan, Cascade, Segmented, Chute, Tiered

a. _____ **horsetail**
b. _____ **segmented**
c. _____ **chute**
d. _____ **cascade**
e. _____ **plunge**
f. _____ **block**
g. _____ **punchbowl**
h. _____ **fan**
i. _____ **tiered**
j. _____ **ribbon**

45. When a porous, water-bearing layer of rock, sand , or gravel is trapped between upper and lower layers of less permeable substrate (often clay) is called a __ (1pt)

- Unconfined aquifer
- **Confined aquifer**
- Watershed
- Zone of aeration

For questions 46, use the diagram below and your knowledge of aquifers.



46. Label the following diagram (1pt each)

Word Bank: aquifer recharge area, confined aquifer, unconfined aquifer, upper confining layer, lower confining layer, ground water, water table

A. _____ **Water Table**

B. _____ **Ground Water**

C. _____ **Unconfined Aquifer**

D. _____ **Confined Aquifer**

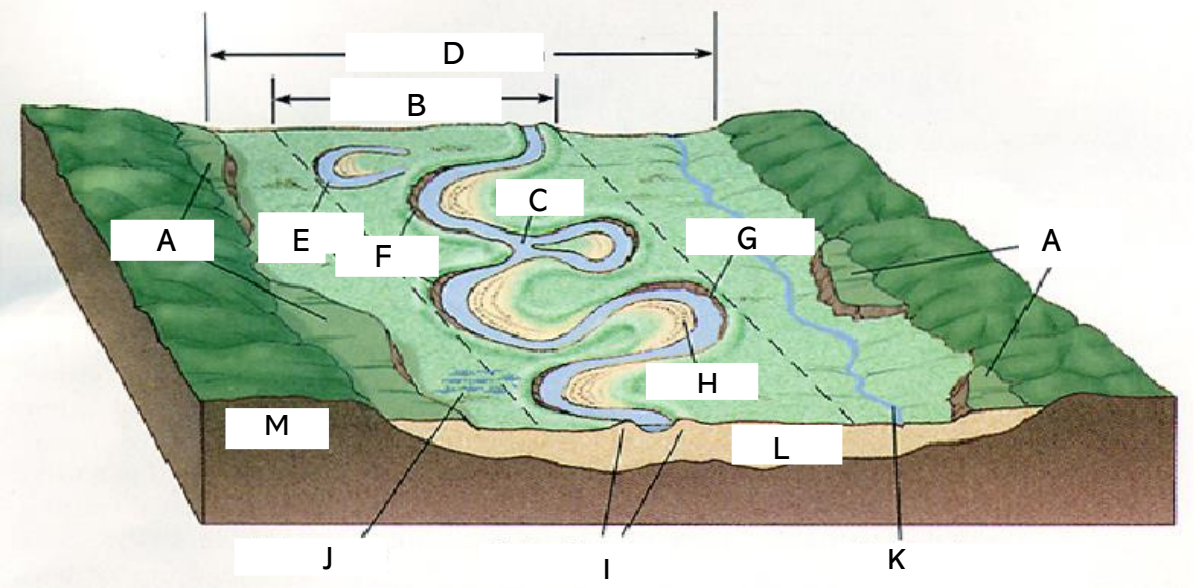
E. _____ **Upper Confining Layer**

F. _____ **Lower Confining Layer**

G. _____ **Aquifer Recharge Area**

47. For question 47, please identify the following floodplain features.

Word bank: Alluvium, Backswamp, Bedrock, Broad floodplain, Cut bank, Cutoff, Meander, Meander belt, Natural leveles, Oxbow lake, Point bar, Stream, Terrace



A. Terrace; B. Meander belt; C. Cutoff; D. Broad floodplain; E. Oxbow lake; F. Cut bank; G. Meander; H. Point bar; I. Natural levees; J. Backswamp; K. Stream; L. Alluvium; M. Bedrock

48. (7 points) A lake has a surface area of 50000m^2 and a water surface elevation of 96.7m relative to a reference datum at the beginning of November. The lake is fed by a surface runoff that has an average inflow of 3.4m/s during the month of November. The lake also has a average outflow in the form of a small stream of 4.5m/s during the month of November. During this month, the lake also received 84mm of rainfall and lost an estimated 5.0 cm height due to evaporation. Calculate the water surface elevation relative to the reference datum at the end of the month of November assuming a constant lake surface area of 50000m^2.

$$(I\Delta t + PA) - (Q\Delta t + EA) = \Delta S$$

Δt = 1 month = 30days X 24hr/days X 60min/hr X 60s/min = 2.592 X 10^6 s
Inflow Volume = IΔt = 3.4m/s X 2.592 x 10^6s= 8.813x10^6 m3
Outflow Volume = QΔt = 4.5m/s X 2.592 x 10^6 s = 1.1664x10^7 m^3
Input due to precipitation = PA = (0.084m X 5x10^8 m^2) = 4.2x10^7 m^3
Output due to evaporation = EA = (0.05m X 5x10^8 m^2)=2.5x10^7 m^3
ΔS = (0.8813+4.2-1.1664-2.5)x10^7 m^3 = 1.415x10^7 m^3
Change in elevation Δh = ΔS/A = (1.415x10^7m^3)/(5x10^8) = 0.0283 m
New water surface elevation at the end of the month = 96.7000m + 0.0283m = 96.7283 m

49. Given an aquifer with a discharge rate of 0.005 m^3/s, at two different injection wells A and B 50m apart, the water table levels were measured to be 50cm and 29cm respectively. Assume the average cross-sectonal area of the aquifer to be $1000m^2$. Find the hydraulic conductivity of the aquifer matrix. Suggest a potential composition of the aquifer matrix (assume uniform and not a mix). (5 pt) **(Use Darcy's Law)**

$$Q = -KiA \rightarrow K = -\frac{Q}{iA} = \frac{0.005\mathrm{m}^3}{s} * \frac{1}{0.0042} * \frac{1}{1000m^2} = 1.19 * 10^{-3} m/s$$

$$i = \frac{dh}{dl} = \frac{0.50m - 0.29m}{50m} = 0.0042$$

Gravel (10^{-4}-10^{-2}m/s) or karst limestone (10^{-6} – 10^{-2} m/s) [answer is in both of these ranges]

50. (2 pts) Describe the formation process for a lake through faulting and rifting.
Faulting and rifting results in an earthquake when a weak section of earth's crust shifts and separates, which can create depressions which form cavities within which rainfall and groundwater can collect to form lakes.

51. (2 pts) Describe at least 2 formation processes for a lake through volcanic action.
As volcanic material is discharged from volcanoes during eruptions, cavities remain which are sometimes unable to drain and thus are filled by runoff from higher elevations, creating a lake. Volcanic damming can also block a stream and create a lake. Explosion craters (called Maars), calderas, cinder cone craters, and lava flow lakes (collapsed lava flow caverns.

52. (3 pts) What is the impact of a levee on water diversion, flooding and the surrounding ecosystem?
Levees increase water speed which increases erosion, harming beneficial in-stream vegetation and increasing downstream flooding. (Answers may vary)

53. (1 pt) What is the primary defining characteristic of generally classified Karst terrain?

Composed of soluble rocks

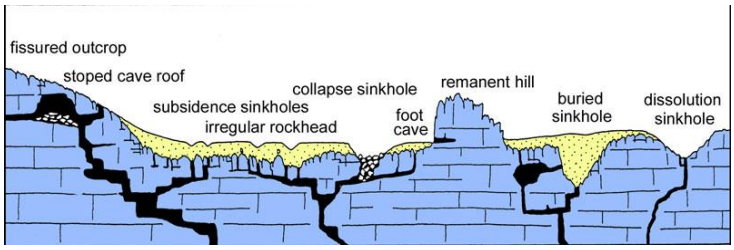
54. (2 pt) What are the two most common types of rock found within Karst terrain

Limestone and dolomite

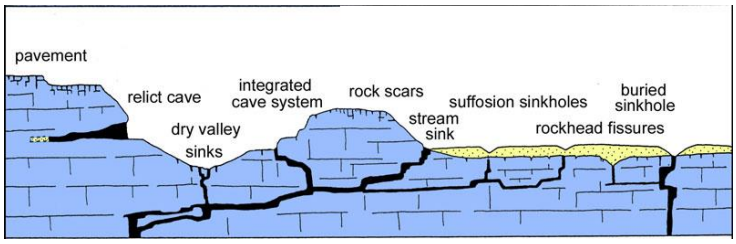
55. (1 pt each) Match each Karst with one of the following descriptors:

Mature, Juvenile, Youthful, Complex, Extreme

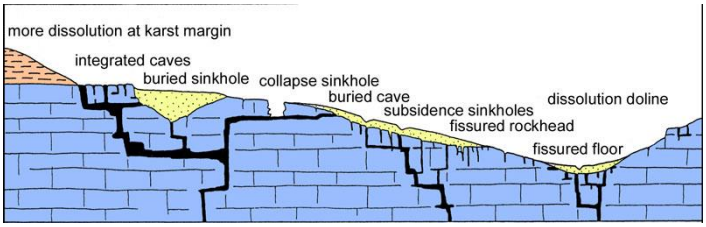
i.____**Complex**_____



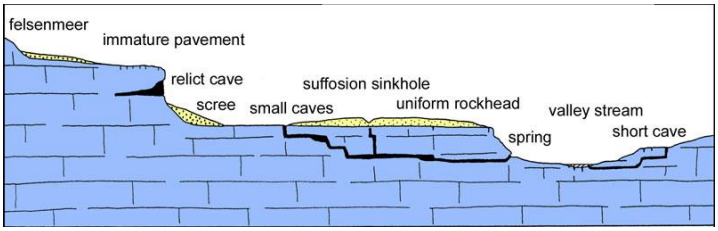
ii.____**Youthful**_____



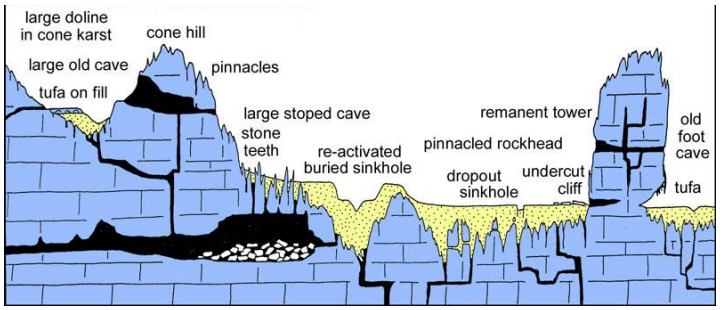
iii._____**Mature**_____



iv. _____**Juvenille**_____



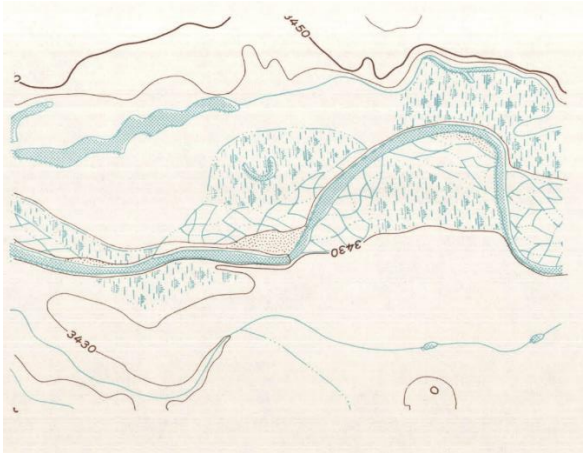
v. _____**Extreme**_____



56. Answer with low or high. Darcy's law is invalid in some ____ permeability materials, with a _____ Reynolds number and _____ hydraulic gradient. (3 pt)

LOW, HIGH, LOW

58. (1 pt each) Examine this map below which is a USGS map depicting the Pecos River in New Mexico with swamps, lakes, streams, and a sandbar. Using knowledge of USGS waterfeature map marking, match what each of the following symbols on the map or in general within hydrographic feature mapping mean to the following features in the word bank. (Note: not all of the symbols/features are exhibited in the example map and not all words will be used)



Word bank*: valleys, perennial water source, waterfalls & rapids in a large stream, waterfalls & rapids in a small stream, spring, topographic lines, well, intermittent drain, gorge, area water feature, wash, river*

a. Dash-dot lines _____ Intermittent drain
b. Blue shading _____ area water feature
c. Brown line_____ topographic line
d. Small circle _____ well
e. Small circle with a pointing tail _____ spring
f. What direction does this tail point?
downstream
g. Hachures _____ waterfalls & rapids in a large stream
h. Crossbars _____ waterfalls & rapids in a small stream
i. Solid lines _____ perennial water source

59. Briefly describe what a crest stage gauge measures and how it does it. (2 pts)
Maximum water level, consists of a metal pipe filled with a cork which rises with water level which fills the pipe through holes in its bottom. The cork sticks to a wood staff as the water climbs. When the water level recedes, the cork line marking the maximum water level remains.

60. What is down-cutting? (1 pt)
Down-cutting is when a river erodes away the stream bed, carving out a canyon or valley; occurs during flooding.

True/False Section

61. Take Tanganyika is the world's largest freshwater body by volume. **(False)**
62. The most abundant cations in freshwater are bicarbonate, chloride, and sulfate. **(False)**
63. There is more freshwater stored in the Antarctic Ice Sheet than anywhere on Earth. **(True)**
64. A micro-watershed can be as small as an acre. **(True)**
65. Nitrates and phosphates are the leading freshwater contamination sources. **(True)**
66. Pingos are formed from rising hydraulic pressure in permafrost gaps causing water to freeze and uplift overlying alluvium. **(True)**
67. The zone of saturation is the area of an unconfined aquifer above the water table where pore space is filled with air. **(False)**
68. Waves in lakes are most commonly created by uneven heating of the lakes surface creating thermal gradients. **(False)**
69. Sewage and waste water treatment is the largest point source of freshwater pollution. **(True)**
70. Permeability is defined as the quality or degree of material of having minute holes through which a fluid may pass. **(False)**
71. Capillarity refers to the action of particulates in a porous material to slow the flow through the material through adhesive, cohesive, and surface tension forces. **(False)**
72. Ephemeral streams are a subclassification of intermittent stream flows. **(False)**
73. Contaminated groundwater is most commonly treated through the pump and treat method. **(True)**
74. Stream discharge is most commonly measured through dilution gauging. **(False)**
75. Evapotranspiration is an outflow process of water budgets. **(True)**
76. Flyoff is another word for evapotranspiration. **(True)**

Pollution Free Response (2 pts each)

77. _____ groundwater pollution is more persistent and difficult to address than _____ surface water pollution. (groundwater / surface water)

78. What is the difference between point source and non-point source pollution?

Point source comes from discrete locations like a factory or sewer pipe, where no-point source is cumulative from multiple inputs over large areas such as farms, city streets, and residential neighborhoods

79. What 1977 Law established the basic structure for regulating discharges of pollutants into the waters of the United States and regulating quality standards for surface waters?

The Clean Water Act

80. A "dead zone" is an area of low _____ levels, which appears annually in the Northern Gulf of Mexico, feuled by nutrient runoff from Midwest farms.

Dissolved oxygen

Bonus Tiebreaker : What is the largest dam in the world and what river does it dam?
The Three Gorges Dam on China's Yangtze River